METRIC
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30 May 1997
SUPERSEDING
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26 May 1992

PERFORMANCE SPECIFICATION

HEATER ASSEMBLY, COMBUSTION VEHICULAR COMPARTMENT, 8.8/17.6 kW (30 000/60 000 BTU/HR) GENERAL SPECIFICATION FOR

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a forced draft and ventilation air personnel heater assembly, referred to as a "heater" (see 6.1).

1.2 Classification. Heaters should be of the following types and classes, as specified:

Type I	- Heaters that must take their combustion and ventilating air from the same source ("Single Air").
Class I	- Heaters that have a nominal rating of 8.8 kilowatts (kW) (30 000 British Thermal Units per Hour [Btu/hr]), but are also capable of operating at a nominal heating rate of 4.4 kW (15 000 Btu/hr).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2540

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Class II	- Heaters that have a nominal maximum rating of 17.6 kW (60 000 Btu/hr), but are also capable of operating at a nominal heating rate of 8.8 kW (30 000 Btu/hr).
Type II	- Heaters that can take their combustion and ventilating air from separate sources (“Dual Air”).
Class II	- Heaters that have a nominal maximum rating of 17.6 kW (60 000 Btu/hr), but are also capable of operating at a nominal heating rate of 8.8 kW (30 000 Btu/hr).

2. APPLICABLE DOCUMENTS.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

A-A-52472	- Filter, Fluid, Pressure, Automotive Fuel (10 GPH Filtration).
MIL-PRF-62550/1(AT)	- Heater Assembly, Combustion, Vehicular Compartment, 17.6 kW, Single Air Type (Metric).
MIL-PRF-62550/2(AT)	- Heater Assembly, Combustion, Vehicular Compartment, 17.6 kW, Dual Air Type (Metric).
MIL-PRF-62550/3(AT)	- Heater Assembly, Combustion, Vehicular Compartment, 8.8 kW, Single Air Type (Metric).

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

11669705	- Box Assy, Control (Interface).
12268109	- Box Assembly, Accessory Control (Interface).
12330264	- Control Box, Electrical, Heat-Ventilator (Interface).
12345539	- Driver's Master Panel Assembly (Interface).

PURCHASE DESCRIPTIONS

ATPD 2165	- Heater Test Stand, Valcom model VHTS-89102.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.)

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Part 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DEPARTMENT OF TRANSPORTATION (DoT)

Code of Federal Regulations (CFR) - Title 49, Part 393.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Performance Test Code 19.5	-	Application Part II of Fluid Meters Sixth Edition 1971 (Interim Supplement 19.5 on Instruments and Apparatus) (DoD Adopted).

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 E. 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117	-	Standard Practice for Operating Salt Spray (Fog) Apparatus (DoD Adopted).
ASTM D975	-	Standard Specification for Diesel Fuel Oils (DoD Adopted).
ASTM D1655	-	Standard Specification for Aviation Turbine Fuels (DoD Adopted).
ASTM D2156	-	Standard Test Method for Smoke Density in Flue Gases from Burning Distillate Fuels (DoD Adopted).

(Application for copies should be addressed to American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AIR MOVEMENT AND CONTROL ASSOCIATION (AMCA)

AMCA 210	-	Laboratory Methods of Testing Fans for Rating (DoD Adopted).

(Application for copies should be addressed to the Air Movement and Control Association, 30 W. University Drive, Arlington Heights, IL 60004.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE J726	-	Air Cleaner Test Code, Recommended Practice (DoD Adopted).

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.4 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for related associated performance specification sheets), the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.d), a sample shall be subjected to first article inspection (see 4.3).

3.2 Design and construction. The heater shall meet the physical (see 3.3), environmental (see 3.4), performance (see 3.5), and reliability/durability (see 3.6), requirements as specified herein. The heater shall be constructed to the form and dimension depicted on the applicable performance specification sheet (see 6.2) and shall be compatible with control box 11669705, 12268109, 12330264, panel assembly 12345539 and test stand ATPD 2165 (see 4.7.1).

3.3 Physical.

3.3.1 Weight. Class I heaters shall not weigh more than 9.5 kilograms (kg) [21 pounds (lb)]. Class II heaters shall not weigh more than 17 kg (38 lb) (see 4.7.3).

3.3.2 Materials. Heaters shall be in accordance with the manufacturer's materials specifications for heaters, and shall meet the requirements specified herein (see 4.7.1).

3.3.2.1 Dissimilar metals. Except where necessary to complete an electrical circuit, dissimilar metals shall not be electro-chemically incompatible (see 4.7.1).

3.3.2.2 Hazardous materials. Asbestos, Cadmium, and radioactive material will not be used in this item. Radioactive material is defined by a) Title 10, Code of Federal Regulations, Part 40, and b) Other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item (see 4.7.1).

3.3.2.3 Electrical components and wiring. The wire and insulation shall be flexible to -54 degrees Celsius (°C) [-65 degrees Fahrenheit (°F)]. Each wire end or connector shall be uniquely and permanently marked. Marking the ignitor leads is optional if they are readily identifiable (see 4.7.1).

3.3.2.4 Brush inspection. All brush-type electric motors shall have provisions for brush inspection or indication of brush wear without disassembly (see 4.7.1).

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3.3.2.5 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.3 Interchangeability. MIL-PRF-62550/2 heaters shall be easily convertible to MIL-PRF-62550/1 through the use of screw drivers, hex keys or hex sockets. The conversion of heaters shall not remove any protective or sensing devices needed to fulfill requirements specified herein. After conversion, dimensional and functional conformance to MIL-PRF-62550/1 shall be verified (see 4.7.1 and 4.7.4).

3.3.4 Welding. All heater welds that are exposed to combustion gases during normal operation shall be readily weldable for highly stressed joints. Welding procedure shall be in accordance with the best commercial standards (see 4.7.1).

3.3.5 Combustion air inlet strength. Type II only. The heater's combustion air inlet shall be capable of sustaining at least 50 kg (110 lb) load without permanent bending, cracking, interference between moving parts, or any damage that would effect heater performance or durability. Combustion air inlet adapter rotating torque increase shall be less than 20 percent (%) at the end of the test (see 4.7.5).

3.3.6 Safety. Surface areas of the heater outside of designated hot spots shall not cause contact burns. Heaters shall comply with all DoT requirements contained in 49CFR 393.77 "Heaters". The heater shall not make more noise than 85 decibels [dB(A)], and shall also not increase the overall noise level in the crew compartment (see 4.7.1 and 4.7.6).

3.3.7 Protective coating. The heater shall be coated with dull olive or forest green paint that will not be damaged by handling or exposure to environmental testing (see 4.7.2 and 4.7.7).

3.3.8 Identification and marking. Nameplate marking shall withstand decontaminating agents (see 3.3.7) and the specified environmental tests (see 3.4). After environmental tests information shall be readily legible from 91.5 centimeters (cm) (3 feet) by a person with 20-20 vision. Identification marking shall include, as a minimum, the following information permanently marked on the exterior of the heater (see 4.7.1 and 4.7.2).

a. Design activity CAGE and Part or Identifying Number (PIN).
b. Manufacturer's name, CAGE, and model number.
c. Contract number.
d. Nomenclature: Heater, Vehicle Compartment, Fuel-Fired, 17.6 kilovolts (kV) (60 000 Btu/hr) or 8.8 kW (30 000 Btu/hr), as applicable.
e. National stock number.
f. Serial number.

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g. Type II (only) heaters. Combustion air inlet adapter shall be marked as follows with letters at least 1.25 cm (0.5 inch (in.)) high by dull black paint: "DO NOT REMOVE".

3.4 Environmental conditions. The heater shall meet the performance requirements of 3.5 and retain its maintainability after being subjected to the environmental conditions specified in 3.4.1 through 3.4.7 (see 4.7.7).

3.4.1 Temperature.

3.4.1.1 High temperature. Storage at a temperature of at least 82°C (180°F) and operation with combustion air temperature of at least 50°C (120°F) (see 4.7.8.1).

3.4.1.2 Low temperature. Storage at -54°C (-65°F) and operation down to -50°C (-55°F) (see 4.7.8.2).

3.4.2 Humidity. Exposure to temperature and humidity cycles (see 4.7.9).

3.4.3 Salt fog. Exposure to salt fog for 96 hours (see 4.7.10).

3.4.4 Shock. Exposure to 40 gravity units (g) shock loading for 18 milliseconds (see 4.7.11).

3.4.5 Vibration. Exposure to sinusoidal vibration in accordance with figure 1, including up to four resonance dwells (see 4.7.12).

3.4.6 Electromagnetic interference (EMI). Operate the heater as specified (see 3.5.7 and 3.5.1.1). For all operating modes (including start cycle) it shall not emit signals on the control box's power lead greater than shown by figures 3a and 3b or radiate signals greater than shown by figures 4 and 5. Also it shall not be damaged or malfunction when exposed to interfering signals shown by table I (radiated) and figure 6 (control box's power lead) (see 4.7.13).

3.4.7 Dust. Exposure for at least 200 hours with at least 3 grams per cubic meter (g/m^3) of fine dust per SAE J726 in the combustion air (see 4.7.14).

TABLE I. Radiated interference.

Frequency Range	Minimum Limit Level (Volts/Meter)
10 kHz to 2 MHz	10
2 MHz to 30 MHz	20
30 MHz to 2 GHz	50
2 GHz to 12.4 GHz	10

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NOTE: Above 30 MHz expose the heater to both vertically and horizontally polarized signals.

3.5 Performance. The heater shall meet the performance requirements specified herein when mounted in any position from horizontal to vertical with the air inlet level with, or higher than, the air outlet; and, except as otherwise specified, when operating in the high, and low heat modes.

3.5.1 Control circuits.

3.5.1.1 Start cycle. The heater control circuits shall control the heater start cycle and, upon starting, automatically switch to run mode and illuminate the control box run indicator. The starting time, measured from the time the control box's "start-off-run" switch is put in the "start" position until, the heater no longer needs the energy input from the ignitor and its circuit is de-energized shall not exceed 3.5 minutes (see 4.7.15).

3.5.1.2 Starting with fault. If the heater fails to start in the allotted time (see 3.5.1.1), the heater shall automatically exit the start mode, diagnose the fault, and exhaust any fuel vapors (see 3.5.13). The heater shall not be damaged when subjected to the conditions described in a, b, and c below. For conditions b and c only, there shall be less than 140 cubic centimeters (cm^3) (class I) or 240 cm^3 (class II) of total fuel, it shall enter its purge cycle, shut off the fuel flow, and then enter its self diagnostic mode in accordance with 3.5.2 (see 4.7.16).

a. Heater control box's "start-off-run" switch in the "start" position for more than 30 minutes.
b. Open ignitor circuit.
c. Shorted ignitor circuit.

3.5.2 Self diagnostics. The heater shall indicate support system and repair part faults by flashing the control box run indicator. Individual faults shall be identified by the heater, and shall be visible and readable from the control box cover, without using any tools onboard, and without the need for reference to additional documentation by maintenance personnel (see 4.7.17).

3.5.3 Fuel.

3.5.3.1 Fuel compatibility. The heater shall be capable of operating on military aircraft fuels, Grade No's 1-D and 2-D per ASTM D975 and Jet A-1 per ASTM D1655 (see 4.7.18).

3.5.3.2 Fuel leakage. There shall be no leakage from the heater fuel components (see 4.7.19).

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3.5.3.3 Low fuel pressure. When exposed to low fuel pressure, the heater shall flash the control box run indicator and shall enter the self diagnostics mode according to 3.5.2. Low fuel pressure shall be detected at less than 27.6 kilopascals (kPa) [4.0 pounds per square inch (lb/in^2) (pounds per square inch gage (psig))] or between 6.9 and 13.8 kPa (1 and 2 psig) above the minimum demonstrated operating fuel pressure, provided the resultant pressure is less than 27.6 kPa (4 psig) (see 4.7.20). Minimum demonstrated operating fuel pressure is the fuel pressure below 20.7 kPa (3.0 psig) which is proven to provide operating capability.

3.5.3.4 Fuel pressure. The heater shall operate with fuel pressures between 20.7 (or lower, see 3.5.3.3) and 103 kPa (3 and 15 psig) (see 4.7.20.1).

3.5.4 Voltage. The heater shall start and meet performance requirements with applied voltages of 23 to 30 volts (V) direct current (dc). At ambient temperatures below -32°C (-25°F) the heater shall start and run with an applied voltage of 19 Vdc (see 4.7.21).

3.5.4.1 Transient voltage. The heater shall not be damaged or malfunction when exposed to electrical spikes, surges and ripple generated by vehicle electrical systems (see 4.7.22).

3.5.5 Current. At 24 Vdc, the current draw from the power source shall not exceed that specified by table II (see 4.7.23).

TABLE II. Maximum current draw.

Ambient temperature		Class I		Class II	
°C	(°F)	Start (A) 1/	Run (A)	Start (A)	Run (A)
Above 7.2	(45)	12.0	8.0	18.0	18.0
Below 7.2	(45)	17.0	13.0	22.5	22.5

1/ A = Amperes.

3.5.6 Power loss. When power loss occurs there shall be less than 140 cm^3 (class I) or 240 cm^3 (class II) of total fuel flow. The heater shall not enter the start cycle or operate until the control box's "start-off-run" switch is placed in the "start" position (see 4.7.24).

3.5.7 Heat output. The heat output of the heater shall be as shown by table III. High and low heat shall be manually setable by the control box (see 4.7.25).

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TABLE III. Heat output.

Control box heat level setting	Class I kW 1/ (Btu/hr) 2/	Class II kW (Btu/hr)
High heat	8.8 + 0.9 - 0.9 (30 000 + 3000 - 3000)	17.6 + 3.6 - 1.8 (60 000 +12 000 - 6000)
Low heat	4.4 + 0.9 - 0.9 (15 000 + 3000 - 3000)	8.8 +1.8 - 0.9 (30 000 + 6000 - 3000)

1/ kW = kilowatt 2/ Btu/hr = British thermal units per hour

3.5.8 Ventilating air discharge rate. With 24 Vdc applied, the ventilation air discharge rate at the restrictions specified in table VIII or free flow shall be as specified by table IV. Test results shall be corrected to standard air (see 4.7.26):

Density = 1.20 kilogram (kg) per cubic meter, (kg/m^3) (0.075 lb per cubic foot (lb/ft^3)
Pressure = 101.3 kPa (29.92 inches of Mercury (in Hg))
Temperature = 20°C (68°F)
Relative Humidity = 50%

TABLE IV. Ventilation and combustion air discharge rates.

Minimum ventilation air flow		Specified air circuit restrictions			
		Class I		Class II	
kl/min 1/	(scfm) 2/	Ventilation	Combustion	Ventilation	Combustion
3.26	(115)	free flow	free flow	N/A	N/A
2.27	(80)	table X	table X	N/A	N/A
(Low heat)					
5.38	(190)	N/A	N/A	free flow	free flow
4.81	(170)	N/A	N/A	table IX	table IX
(High heat)					
6.37	(225)	N/A	N/A	free flow	free flow
5.82	(205)	N/A	N/A	table IX	table IX

1/ kl/min = kiloliter per minute. 2/ scfm = standard cubic feet per minute.

3.5.8.1 Ventilation. Class II heaters only. When the heater is used in the ventilation mode the ventilation circuit shall meet the air flow rate as specified in 3.5.8. The air flow rate in the combustion circuit shall not exceed 28 milliliters per minute (ml/min) (0.001 scfm) (see 4.7.27).

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3.5.8.2 Seal leakage. When Class II heater is attached to a duct with a diameter between 202 and 208 millimeters (mm) (7.94 and 8.25 inches) and operated on high heat the ventilation air leakage shall be less than 85 liters per minute (1/min) (3 scfm) (see 4.7.28).

3.5.8.3 Ventilating air restriction. The heater shall limit the discharge air temperature between 150 and 177°C (300 and 350°F). When the air flow is restricted to less than 1.27 ± 0.42 kl/min (45 ± 15 scfm) (class 1) or 2.55 ± 0.85 kl/min (90 ± 30 scfm) (class II) the ventilating air temperature shall not exceed 177°C (350°F), the heater shall shut off its fuel flow, enter its purge cycle, and then shut itself off (see 4.7.29).

3.5.9 Inlet over pressure. Class II heaters shall start, operate and have a combustion air flow of not more than 623 l/min (35 scfm) and meet the requirements of 3.5.7 when the inlet of the combustion air circuit is subjected to at least 2.0 kPa (8.0 in. H_20) pressure (see 4.7.30).

3.5.10 Combustion air flow. Class II heaters shall experience no damage or cause a hazardous situation when subjected to the combustion air flow conditions described below. Upon occurrence of these conditions, the heater shall shut off its fuel flow and enter the self diagnostic mode according to 3.5.2 (see 4.7.31). The exhaust smoke spot number throughout shall be less than three per ASTM D2156.

3.5.10.1 Reverse flow. The heater shall be subjected to a reverse pressure of at least 1.5 kPa (6.0 in H_2O).

3.5.10.2 Reduced flow. The heater shall be subjected to a reduced combustion air flow of not less than 280 l/min (10 scfm).

3.5.11 Combustion exhaust. Combustion exhaust temperature shall not drop below 260°C (500°F) (see 4.7.32).

3.5.12 Smoke density. After the heater has been operating for not more than 7 minutes, its exhaust gases shall not exceed a smoke spot number of three per ASTM D2156 (see 4.7.33).

3.5.13 Purging cycle. The purging time shall not exceed 4.0 minutes and the exhaust temperature shall not be greater than 177°C (350°F) when the blower shuts off (see 4.7.34).

3.5.14 Toxic fumes. New heat exchangers shall not leak. After completion of environment and durability testing the heat exchangers' leakage shall not exceed 14 L/min (0.5 cubic foot per minute (ft^3/min)) and shall have no cracks (see 4.7.35).

3.6 Reliability and durability. During the reliability and durability verification 4.7.36 and testing 4.7.37, the heater shall operate on contaminated diesel fuel. During testing the heater shall meet the requirements of 3.5.1.1, 3.5.4, 3.5.5, 3.5.10.2 and 3.5.13 and run with transient voltages

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down to 19 Vdc. After completion of each phase the heater shall meet the requirements of 3.5.1.1, 3.5.3.3, 3.5.4, 3.5.5, 3.5.7, 3.5.10.2, 3.5.12, and 3.5.13 (see 4.7.36).

3.6.1 Reliability. The heater shall have a minimum reliability of 1250 mean hours between failure (MHBF). A reliability failure is defined as any malfunction of the heater that requires component replacement above unit level (see 4.7.37).

3.6.2 Durability. The heater shall have a minimum durability requirement of 2500 operating hours with at least 2400 start cycles without failure. A durability failure is defined as any malfunction of the heater that is not repairable at direct support (see 4.7.37).

3.6.2.1 Reduced durability. For control test only, the durability testing shall be reduced to 300 cycles (see 4.7.37.1).

4. VERIFICATION

4.1 Inspection equipment. Inspection equipment shall be capable of repetitive measurements with an error less than 10% of the parameter's tolerance. When a requirement must be calculated from measurements the error is associated with the requirement and the contractor is responsible for choosing instrumentation with errors that will produce the specified error (calculated) in the requirement. None toleranced parameters shall be measured with a maximum error of 5% of reading. All measurements and calibrations shall be traceable to the National Institute of Technology and Standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).
c. Control tests (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all inspections shall be conducted under the following conditions:

a. Air temperature: 23 $\pm$ 10°C (73 $\pm$ 18°F).
b. Barometric pressure: 96.5 $\pm$ 6.8 kPa (28.5 $\pm$ 2 in Hg).
c. Relative humidity: 10 to 90%.

4.3 First article inspection. The first article test plan shall be approved by the procuring activity prior to beginning of first article testing. The Government shall randomly select six of the first 25 production heaters produced under the production contract for first article inspection.

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First article samples shall be numbered and inspected as specified in 4.3.1 and 4.3.2. Unless otherwise specified, the listing in table V does not imply any particular test sequence.

4.3.1 In-process inspections. During fabrication of the first sample production heater, an in-process inspection shall be conducted by the contractor under the surveillance of the Government inspector. Inspection shall be made prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records shall be reviewed and evaluated during this inspection.

4.3.2 Production sample. After completion of inspection, the first sample production heater shall remain at the manufacturing facility, as a production sample, and be the last heater shipped on the contract. The heaters subject to durability testing shall not be used to fulfill production quantities of the contract.

4.4 Conformance inspections (CI). CI shall consist of the examinations of 4.4.2 and the acceptance tests of 4.4.3. Nonconformance with any of the specified requirements in sections 3 and 5 shall be cause for rejection.

4.4.1 Lot formation. An inspection lot shall consist of all the heaters of one type and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.4.1.1 Sampling for inspection. The contractor shall use statistical quality management to select sample size necessary to ensure a six sigma quality level. Before sampling may be initiated, the contractor shall inspect 100% of the first 100 heaters to establish that the process average (see 6.6) falls within the control limits.

TABLE V. Classification of inspections.

Title	Requirement	Inspection	First article test		Conformance	
			1	2	Acceptance 100%	Control
			Hor 1/	Vert 2/	Hor	Hor
Materials, design & construction	3.2, 3.3.2, 3.3.3, 3.3.4, 3.3.6, & 3.3.8	4.7.1	X	X		X
Defects	3.2, 3.3.7, & 3.3.8	4.7.2	X	X		X
Weight	3.3.1	4.7.3	X	X		
Interchangeability	3.3.3	4.7.4	X	X		
Combustion air inlet strength	3.3.5	4.7.5	X	X		
Noise level	3.3.6	4.7.6	X	X		

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TABLE V. Classification of inspections - Continued.

Title	Requirement	Inspection	First article test		Conformance	
			1	2	Acceptance 100%	Control
			Hor 1/	Vert 2/	Hor	Hor
Environmental conditions	3.4	4.7.8	X	X		
High temperature	3.4.1.1	4.7.8.1	X	X		
Low temperature	3.4.1.2	4.7.8.2	X	X		
Humidity	3.4.2	4.7.9	X	X		
Salt fog	3.4.3	4.7.10	X	X		
Shock	3.4.4	4.7.11	X	X		X
Vibration	3.4.5	4.7.12	X	X		X
Electromagnetic interference	3.4.6	4.7.13	X	X		
Dust	3.4.7	4.7.14	X	X		
Starting cycle	3.5.1.1	4.7.15	X	X	X	X
Starting with fault	3.5.1.2	4.7.16	X	X		
Self diagnostics	3.5.2	4.7.17	X	X		
Fuel compatibility	3.5.3.1	4.7.18	X	X		
Fuel leakage	3.5.3.2	4.7.19	X	X	X	X
Low fuel pressure	3.5.3.3	4.7.20	X	X		X
Fuel pressure	3.5.3.4	4.7.20.1	X	X		X
Voltage	3.5.4	4.7.21	X	X	X	X
Transient voltage	3.5.4.1	4.7.22	X	X		
Current	3.5.5	4.7.23	X	X	X	X
Power loss	3.5.6	4.7.24	X	X	X	X
Heat output	3.5.7	4.7.25	X	X	X	X
Ventilating air discharge rate	3.5.8	4.7.26	X	X	X	X
Ventilation	3.5.8.1	4.7.27	X	X		
Seal leakage	3.5.8.2	4.7.28	X	X		X
Ventilation air restriction	3.5.8.3	4.7.29	X	X	X	X
Inlet over pressure	3.5.9	4.7.30	X	X		
Combustion air flow	3.5.10	4.7.31	X	X		
Combustion exhaust	3.5.11	4.7.32	X	X		
Smoke density	3.5.12	4.7.33	X	X		
Purging cycle	3.5.13	4.7.34	X	X	X	X
Toxic fumes	3.5.14	4.7.35	X	X	X	X
Reliability & durability	3.6	4.7.36 4.7.37	X	X		
Reduced durability	3.6.2.1	4.7.37.1				X

1/ = Horizontal
2/ = Vertical

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4.4.2 Examination. Sample units shall be examined for the defects specified in table VI.

TABLE VI. Classification of defects.

Category	Defect	Method of examination
Major		
101	Assembly, incomplete (see 3.1).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.1).	SIE 1/
Minor		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.1).	SIE
202	Protective coating, improper (see 3.3.7).	Visual
203	Identification marking, improper (see 3.3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.4.3 Acceptance test. Acceptance test shall be conducted in accordance with the tests specified in table V. Each heater shall be 100% inspected for each requirement. Any defect shall be cause for the Government to refuse acceptance until action taken by the contractor to correct defects and cause has been approved by the procuring agency. Acceptance testing shall be conducted at 23.0 ± 1.0 Vdc and high heat, control box optional.

4.5 Control tests. Control tests shall be conducted at a rate of one out of 600 heaters consecutively produced, except that not more than one test shall be performed in a three month period, nor less than one test in a six month period. Heaters shall be subjected to the control tests specified in table V. Failure of any heater to pass any of the specified tests shall be cause for the Government to refuse acceptance of the quantity represented until action is taken by the contractor to correct defects and to prevent recurrence has been approved by the procuring agency.

4.6 Inspection of packaging. The preservation, packaging, packing, and marking shall be inspected to determine conformance to 5.1.

4.7 Methods of inspection. Unless otherwise specified tests shall be conducted as specified in table V, at the ambient conditions specified in 4.2 at the specified heat outputs, high and low heat (see 3.5.7), and at input voltages (measured at the power input to the control box) of 23.0 ± 0.5 Vdc and 30.0 ± 0.5 Vdc. The fuel pressure shall be 55 ± 7 kPa (8 ± 1 psig). Temperature sensor shall be shielded from radiant heating by the heater. All tests shall be conducted with heater control box part number 11669705 with a nonshielded copper wire cable assembly 3 to 3.5 meters (10 to 12 feet (ft)) long connecting the control box with the heater. Wire gauge shall be equal or greater to that used in the control box for the same circuit. If the test set up requires a longer cable, the gauge of the wire and cable length shall be changed so as to

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provide equal resistance to that of the specified cable. No repairs, adjustments or maintenance may be performed upon the heater during the testing specified herein. When control box run indicator flashing is specified, the heater shall be monitored for identification of specific support system fault causing the flashing (see 3.5.2).

4.7.1 Materials, design and construction. To determine conformance to 3.2, 3.3.2.1 through 3.3.4, 3.3.6, and 3.3.8, inspect contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, control standards, vender catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.2, 3.3.7, and 3.3.8 shall be determined by examination for the defects listed in table VI.

4.7.3 Weight. To determine conformance to 3.3.1, weigh the heater.

4.7.4 Interchangeability. To determine conformance to 3.3.3 remove the dual air adapters using the specified tools. Subject the heater to the tests of 4.7.15, 4.7.25, 4.7.26, 4.7.29, 4.7.33, and 4.7.34. Reinstall the adapter using the same tools and retest the heater as above.

4.7.5 Combustion air inlet strength. Type II, MIL-PRF-62550/2 only. To determine conformance to 3.3.5, the heater shall be suspended with a suitable strap [less than 2.5 cm (1.0 in. wide] from each of the inlet tubes listed below. Weights shall be attached to the heater so that the total weight supported by the specified diameter (heater plus weights) is at least 50 kg (110 lb). Examine the heater for evidence of interfering parts, such as vibration, slow motor speeds, or sounds of rubbing parts. The heater shall be tested at the following combustion air inlet diameters:

a. 38.10 mm (1.50 in.) diameter. Measure the torque necessary to rotate the combustion air inlet through the specified range on the applicable performance specification sheet. Suspend the heater as specified above. Remove the heater from the test setup. Inspect the heater and measure the torque necessary to rotate the combustion air inlet.
b. 44.00 mm (1.7 in.) max. diameter. Support heater from center (within 15%) of tube as specified.

4.7.6 Noise level. To determine conformance to 3.3.6, operate the heater, survey the sound level produced by the heater at a distance of less than one meter (39 in.), locate the noisiest spot, and report on this value.

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4.7.7 Environmental conditions. To determine conformance to 3.4, after exposing the heater to the tests specified in 4.7.8 through 4.7.13, test the heater according to low fuel pressure, fuel leakage, heat output, ventilation air restriction, smoke density, purge cycle, ventilation air discharge rate, seal leakage, current, voltage, start cycle, and combustion air flow. After the final toxic fumes test (see 4.7.35) disassemble the heater with common tools (see 3.3.3). Examine painted surfaces for rust spots greater than 0.5 mm^2 and the parts for damage that makes them unsuitable for their use as repair parts by the applicable military repair level.

4.7.8 Temperature.

4.7.8.1 High temperature. To determine conformance to storage requirements of 3.4.1.1 expose the heater for at least 48 hours to an ambient air temperature not less than 82°C (180°F) in the position specified in table V. After storage reduce the ambient temperature to 23 ± 10°C (73 ± 18°F) and test the heater as specified in 4.7.15, 4.7.19, 4.7.20.1, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.29, 4.7.33, 4.7.34 using Grade No. 2-D fuel. To determine conformance to combustion air requirements of 3.4.1.1 measure the smoke density as specified in ASTM D2156 after operating the heater for a minimum of one hour at high heat using combustion air at a minimum of 49°C (120°F).

4.7.8.2 Low temperature. To determine conformance to 3.4.1.2, preconditioned the heater by operating with Jet A-1 fuel. Store the heater at an ambient temperature below -54°C (-65°F) for at least 12 hours in the position specified in table V. After storage, stabilize the heater and supply Jet A-1 fuel, both at an ambient temperature of -48 ± 3°C (-55 ± 5°F), and test the heater as specified in 4.7.15, 4.7.19, 4.7.20.1, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.29, 4.7.33, 4.7.34. Repeat the test using Grade No. 2-D fuel and a temperature of -4 ± 3°C (25 ± 5°F).

4.7.9 Humidity. To determine conformance to 3.4.2, operate the heater on high heat for at least one hour before exposure to ten cycles of humidity and temperature as shown by figure 2. In less than two hours after completion of the final humidity cycle, the heater shall be tested for the following tests as specified in 4.7.15, 4.7.17, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.33, and 4.7.34. Water drops may be wiped off external heater surface. No other drying is allowed.

4.7.10 Salt fog. To determine conformance to 3.4.3 test the heater in accordance with ASTM B117. In less than two hours after completion of the salt fog exposure, the heater shall be tested for the following tests as specified in 4.7.15, 4.7.17, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.33, and 4.7.34. Water/salt solution drops may be wiped off external heater surface. No other drying is allowed.

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4.7.11 Shock. To determine conformance to 3.4.4, with no fuel applied, expose the heater to a half sine shock pulse with an amplitude of 40 g's and a duration of 18 milliseconds. Apply both negative and positive pulses to each of its three orthogonal axes for a total of six shocks. Mount the heater at the strap space locations shown on the applicable performance specification sheets.

4.7.12 Vibration. To determine conformance to 3.4.5 rigidly mount the heater at the strap space locations shown on the applicable performance specification sheet and subject it to sinusoidal vibration shown by Figure 1. During testing the heater shall be started, purged, operated and monitored for intermittent operation or erroneous indication of faults. Conduct a resonant frequency search along each axis over a range of 5 to 500 hertz (Hz). Include 13 -1/3 minutes dwell time at each resonance. If more than four resonances are present use the most severe. Vary the frequency logarithmically. Cycle for 80 minutes on each axis including the resonance dwell time. Divide cycle time equally between operation at low and high heat.

4.7.13 Electromagnetic interference (EMI). To determine conformance to 3.4.6, perform the following steps as specified herein. Frequency measurements shall be within 0.1% of reading. Radio frequency amplitudes shall be within 2.0 dB:

a. The heater shall installed with mounting straps on a metallic ground plane that is at least 2.25 square meters (m^2) (3487.5 square inch ($in.^2$)). The heater shall be positioned so that it is 10 $\pm$ 2 cm (3.93 $\pm$ .79 in.) from the ground plane's edge and that maximum radiating and sensitive surfaces face the test antennas. During radiated testing the ground plane shall be 80 - 90 cm (31.5 - 35.4 in.) above the test cell's floor and the antenna shall be one meter away from the heater. Base of the rod antenna shall be at the same height of the ground plane. Other antennas' centers shall be 120 cm (47.25 in.) above the test cell's floor. The ground plane resistance shall be less than 0.1 milliohms. The power supply shall be isolated from the heater with a Line Stabilization Network shown in figures 7 and 8. The heater test cable assembly (including heater grounding cable) shall run for at least two meters parallel and along the ground plane's front edge. Each cable shall be 5 cm (1.97 in.) above the ground plane separated by 2 cm (0.78 in.) starting at 10 cm (3.94 in.) from the ground plane and moving in. Each face of the heater shall be probed to determine the area producing the maximum emission or susceptibility. Faces and frequencies known to be most responsive shall be examined. Otherwise probing shall be conducted at least one frequency for each two octave interval in the applicable bandwidth. Heater shall be originated to produce maximum response.
b. A peak detector shall be used for these measurements.

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c. The heater shall be tested while operating as specified by 4.7.15, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.33, and 4.7.34. During susceptibility testing the heater shall be monitored for any malfunctions including erroneous indication of a fault and smoke spot shall not exceed three.

4.7.14 Dust. To determine conformance to 3.4.7, operate a class II heater at 28.0 ± 0.5 Vdc at both heats for at least 100 hours at each heat. During operation fine dust shall be added to the heater's combustion air at the specified rate. Test the heater according to 4.7.25 upon completion of the dust testing.

4.7.15 Starting cycle. To determine conformance to 3.5.1.1 place the control box's "start-off-run" switch in the "start" position for 2 ± 1 seconds. Return the switch to its "run" position. Monitor the control box run indicator. Repeat this test except keep the switch in the "start" position until the control box run indicator lights and then place the switch in the "run" position.

4.7.16 Starting with fault. To determine conformance to 3.5.1.2, the heater shall be subjected to the tests a, b, c, and d below. These tests shall be completed prior to durability and shock testing.

a. Move the control box's start-off-run switch to the "start" position and hold it there for at least 30 minutes. Start and operate the heater.
b. Remove the igniter from the heater and leave the heater's igniter circuit open. Move the control box's start-off-run switch to the "start" position and hold for 30 minutes. Measure and record the total fuel flow delivered to the heater.
c. Repeat the test from item b with the igniter circuit shorted out using a copper wire of at least 10 gauge and not longer than 30 cm (10 in.). Measure and record the total fuel flow to the heater.
d. Re-install the igniter and operate the heater.

4.7.17 Self diagnostics. To determine conformance to 3.5.2 monitor control box run indicator and induce system support and repair part failures. Examine heater for identification of fault.

4.7.18 Fuel compatibility. To determine conformance to 3.5.3.1 perform the durability test as specified in 4.7.37.

4.7.19 Fuel leakage. To determine conformance to 3.5.3.2 operate the heater for at least 5 minutes with a fuel pressure of 96.5 ± 6.7 kPa (14.0 ± 1 psig) applied to the fuel valve inlet. Subsequently remove the heater control box cover and examine the fuel circuit for fuel leakage.

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4.7.20 Low fuel pressure. To determine conformance to 3.5.3.3, perform the following two steps:

Step 1. Before entering the start cycle, lower the fuel pressure at the fuel inlet to 27.6 ± 0.7 kPa (4.0 ± 0.1 psig), or between 6.9 and 13.9 kPa (1.0 and 2.0 psig) above the minimum demonstrated operating pressure, whichever is lower. Start the heater, and monitor the control box run indicator. Shut the heater off. Raise the fuel pressure to 55 + 7 kPa (8 + 1 psig), start the heater, and monitor the control box run indicator.

Step 2. Repeat Step 1, except operate the heater for one hour prior to start of test.

4.7.20.1 Fuel pressure. To determine conformance to 3.5.3.4, operate the heater with fuel pressures between 20.7 (or lower, see 4.7.20) and 103 kPa (3 and 15 psig).

4.7.21 Voltage. To determine conformance to 3.5.4 measure the applied voltage.

4.7.22 Transient voltage. To determine conformance to 3.5.4.1, isolate as needed, heater's power supply from the test signals and subject the heater to the following operations:

a. Operate the heater at 28.0 ± 0.5 Vdc at high heat. Impose a ripple of at least 7 volts peak to peak to heater's power and linearly (±10%) sweep the fundamental frequency over a period of at least one hour between 20 and 200 Hz. Continuously monitor the heater's operation (including the control box's lamp for erroneous flashing) and exhaust gases to determine satisfactory operation and that smoke spot does not exceed three.
b. While the heater is operating apply at least one hundred + and - 250 volt spikes of at least 15 millijoules to the heater's power. The rate of application shall ensure at least 50 spikes shall occur during the off state of any switching transistors.
c. While the heater is operating the + and - voltage surges shown by table VII shall be imposed at least ten times on the heater's power supply:

TABLE VII. Voltage.

Voltage (volts)	Source Impedance (milli Ω)	Time (milliseconds)
40	20	50
100	500	50

4.7.23 Current. To determine conformance to 3.5.5 measure the heater current draw.

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4.7.24 Power loss. To determine conformance to 3.5.6, operate the heater for at least 95 minutes while monitoring the fuel flow. Remove the power from the heater control box. Measure the amount of fuel that flowed after power loss. Wait at least one hour. Restore power to the control box and move the "start-off-run" switch to the "start" position. Repeat this test 10 times.

4.7.25 Heat output. To determine conformance to 3.5.7, perform the following:

a. Assemble test setup as shown in figure 9.
b. Set restrictions and measure air flow per 4.7.26.
c. Measure input and output ventilation air temperatures.
d. Run the heater until the ventilating outlet air temperature has stabilized.
e. When testing at ambient temperatures above 13°C (55°F) and the ventilating air outlet temperature is being limited by the heater's controls, the controls may be locked out if the ventilation outlet temperature is between 135 and 177°C (275 and 350°F). When testing a heater at 23 ± 10°C (73 ± 18°F) that is offered as having a minimum demonstrated operating fuel pressure (see 3.5.3.3), this test shall be performed at the lower pressure.
f. Calculate the heat output using the following equation:

$H = V \times d \times Cp \times T$

H = Heat output in Watts (Btu/hr).
V = Ventilating air flow rate in cubic meter per second (m^3/sec) [cubic feet per hour (ft^3/hr)].
d = Density of ventilating air in kilograms per cubic meter (kg/m^3) [pounds per cubic foot (lb/ft^3)].
Cp = Average specific heat of the ventilating air (from temperature rise across heater) in kilojoules per kilogram per degree C (kj/kg °C) (Btu per lb per °F), including moisture content.
T = Ventilating air temperature rise across heater [outlet temperature, in °C (°F), minus inlet temperature, in °C (°F)].

4.7.26 Ventilating air discharge rate. To determine conformance to 3.5.8 perform the following:

a. Assemble heater test setup per figure 9.
b. Set restrictions as specified in table VIII.
c. Measure the air flow and pressure flow losses in accordance with AMCA 210-74 or ASME PTC 19.5.
d. During subsequent testing no changes shall be made in these restrictions.

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e. The contractor shall verify the coefficient of heat transfer of the ventilating air duct wall is less than 1.14 watts per square meter per °C (0.20 Btu/hr per square foot per °F).
f. Operate the heater and measure the air flow.

TABLE VIII. Air circuit restrictions.

Class	Minimum pressure loss kPa	(in H_2O)	Air flow kl/min	(scfm)
Ventilation				
I	0.19	(0.75)	2.7 ± 0.14	(94 ± 5)
II	0.32	(1.30)	4.9 ± 0.14	(174 ± 5)
Combustion				
I	0.19	(0.75)	0.470 ± 0.014	(16.6 ± .5)
II	0.24	(0.95)	0.832 ± 0.014	(29.5 ± .5)

4.7.27 Ventilation. Class II heaters only. To determine conformance to 3.5.8.1 measure air flow in the combustion and ventilation circuits in accordance with AMCA 210-74 or ASME PTC 19.5 with power applied to pin E of the heaters connector.

4.7.28 Seal leakage. Class II heaters only. To determine conformance to 3.5.8.2 assemble heater test set up as shown in figure 10. Install the heater's ventilation air outlet in a duct with a maximum diameter of 201 mm (7.94 in.) and operate the heater on high heat with the specified restriction of 4.7.26. Collect air leaking past the clearance between the heater and the duct in a plenum sealed between them. Measure the plenum's pressure and hold it within 0.012 kPa (0.05 in. H_20) of the heater's ventilation air inlet. Measure the air flow leaving the plenum in accordance with AMCA 210-74 or ASME PTC 19.5. Repeat the test with an air duct having a minimum diameter of 209 mm (8.20 in.).

4.7.29 Ventilating air restriction. To determine conformance to 3.5.8.3, measure (AMCA 210-74 or ASME PTC 19.5) the ventilating air flow in the duct, its temperature and heater fuel flow. Operate the heater at high heat into the duct used in 4.7.26, restricting the ventilating air flow to not less than 1.68 kl/min (60 scfm) (class I) or 3.36 kl/min (120 scfm) (class II) and until the heater limits the air's temperature between 150 and 177°C (300 and 350°F). Continue to operate the heater for at least an additional 30 minutes. As applicable, within a period of not less than five minutes, continuously reduce the air flow to not more than 1.70 kl/min (60 scfm) (class I) or 3.40 kl/min (120 scfm) (class II). Air temperature shall remain within the specified limits. Continue to operate the heater for at least an additional 30 minutes. Then reduce the air flow to greater than 0.42 kl/min (15 scfm) (class I) or 0.85 kl/min (30 scfm) (class II). Note that the heater shuts off, stops fuel flow and enter its purge cycle without the ventilating air temperature exceeding 177°C (350°F).

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4.7.30 Inlet over pressure. Class II heaters only. To determine conformance to 3.5.9 apply 2.0 kPa (8 in. H_2O) pressure to the heater's combustion air inlet. Measure the combustion air flow in accordance with AMCA 210 or ASME PTC 19.5. Test the heater as specified in 4.7.15 and 4.7.25.

4.7.31 Combustion air flow. Class II heaters only. To determine conformance to 3.5.10, the heater shall be subjected to the following tests (4.7.31.1 and 4.7.31.2) prior to beginning the endurance and reliability test.

4.7.31.1 Reverse flow. Perform steps a and b three times each.

a. Apply air at a pressure at least 1.5 kPa (6.0 in. H_2O) to the heater's exhaust. Combustion exhaust pressure reversal shall not take more than 2.0 seconds. Attempt to start the heater. Monitor the control box run indicator.
b. Operate the heater and apply air at a pressure not less than 1.5 kPa (6.0 in. H_20) to the heaters exhaust for 45 to 60 seconds. Remove the pressure from the exhaust for 3 to 4 minutes. Reapply the pressure to the exhaust for at least another 5 minutes. Monitor the fuel flow, the control box run indicator and combustion air inlet temperature. Measure the combustion air inlet temperature within 3 cm (1.2 in.) of the inlet. The air temperature shall be less than 177°C (350°F). Record the air inlet temperature and the time it takes the heater to enter the purge cycle.

4.7.31.2 Reduced flow. Perform steps a and b three times.

a. Reduce combustion gases flow to less than 280 l/min (10 scfm) measured per AMCA 210-74 or ASME PTC 19.5. Attempt to start the heater. Note if the heater starts and the control box run indicator condition.
b. Operate the heater for at least an hour. Over a period of at least an hour linearly (±10%) reduce the air flow to less than 280 1/min (10 scfm) and measure it as specified by AMCA 210-74 or ASME 9.5. During flow reduction, measure the smoke dot (per ASTM D2156) and monitor the control box run indicator. Record these conditions and the time to enter and the time the heater is in the purge cycle.

4.7.32 Combustion exhaust. To determine conformance to 3.5.11 measure the heater combustion exhaust temperature.

4.7.33 Smoke density. To determine conformance to 3.5.12 measure the smoke density of the exhaust gasses in accordance with ASTM D2156 within seven minutes of ignition.

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4.7.34 Purging cycle. To determine conformance to 3.5.13 stabilize the ventilating air temperature at low or high heat, as applicable. Place the control box's "start-off-run" switch in the "off" position. Monitor and record the time for the exhaust temperature to drop to 177°C (350°F) and for the heater blower to automatically shut off.

4.7.35 Toxic fumes. To determine conformance to 3.5.14, prior to heater assembly, all heat exchangers shall be tested by blocking purpose openings, submerging them under water and apply 6 psig pressure internally for at least 60 seconds. No bubbles shall form on the heat exchanger's surface. As the last test of first article and control tests, pressurize the exhaust tube with air at not less than 0.75 kPa (3 in. H_2O) pressure, block the purpose openings, and measure the air flow rate in accordance with AMCA 210 or ASME PTC 19.5. The heat exchanger shall be examined for cracks visually and fluorescent or contrast dye penetrant inspection.

4.7.36 Reliability and durability verification. To verify the heater assembly meets the reliability and durability requirements of 3.6, two (2) heaters shall be placed on test for 2500 hours each. There shall be no more than four (4) chargeable reliability failures (cumulative) and zero (0) chargeable durability failures on the test samples.

4.7.37 Reliability and durability testing. To determine conformance to 3.6, operate the heater, as specified by paragraphs a through g, for the four phases listed below.

a. Monitor self-diagnostics throughout.
b. Accumulate total operation hours of only cycle "on" time specified in table IX. Shut the heater off for at least ten minutes after each "on" cycle.
c. Maintain fuel at the ambient temperature specified in each phase.
d. Operate the heater with restrictions in 4.7.25 and 4.7.26.
e. Install a filter conforming to A-A-52472 in the fuel line leading to the heater. Verify fuel pressure of 8 ± 1 psig at filter inlet. Add fine contaminate per SAE J726 to the fuel up stream of the filter at a rate less than 0.5 grams per hour. Add 100 grams of contaminate in each of the first three phases (300 g total). When the control box run indicator flashes and the heater enters the self diagnostic mode, replace the filter and continue adding contaminate. (NOTE: Replacement of this clogged filter does not constitute a heater failure.)
f. Verify tests 4.7.15, 4.7.21, 4.7.23, 4.7.32, and 4.7.34, during start and operation of each cycle. Drop the applied voltage (4.7.21) to 19 ± 0.5 Vdc for 2.5 ± 0.5 seconds for every 25th start cycle in phases I and II.
g. Verify tests 4.7.15, 4.7.19, 4.7.20.1, 4.7.21, 4.7.23, 4.7.25, 4.7.26, 4.7.32, 4.7.33, and 4.7.34 after each phase. Do not add contaminant.

PHASE I - At an ambient temperature of 23 ± 10°C (73 ± 18°F) use Grade No. 2-D fuel and perform steps a through d.

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Step a: Set heater at low heat and operate for 25 cycles with an applied voltage of 23 ± 0.5 Vdc.
Step b: Repeat step a using an applied voltage of 28.0 ± 0.5 Vdc.
Step c: Set heater at high heat and repeat steps a and b.
Step d: Repeat steps a through c six times (600 total cycles).

PHASE II - At an ambient temperature of -32.0 ± 3°C (-25 ± 5°F) and using Grade No. 1-D fuel repeat steps a through d three times (300 total cycles), and substitute 30.0 ± 0.5 Vdc for 28.0 ± 0.5 Vdc.

PHASE III - At an ambient temperature of -48.0 ± 3°C (-55 ± 5°F) and using Jet A-1 fuel repeat steps a through d three times (300 total cycles), substitute "less than 19 Vdc" in steps a, and 30 ± 0.5 Vdc in step b.

PHASE IV - Repeat Phases I through III a second time (2400 total cycles).

TABLE IX. Durability test times for class I and II heaters.

Heater class	Phase	Cycles (total)	Cycles on time (minutes)	Operational hours total
I	I	1200	30	600
I	II	600	30	300
I	III	600	30	300
II	I	1200	95	1900
II	II	600	30	300
II	III	600	30	300

4.7.37.1 Reduced durability. For control test only. The test shall consist of 100 cycles each of phases I, II and III (see 4.7.37). Addition of dust not required.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Application information. The heater furnished under this specification is intended to supply warm air to the crew compartment of military vehicles.

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of performance specification sheet (see 3.2). Heaters conforming to type I, class II or MIL-PRF-62550/1 should not be ordered. In their lieu, heaters conforming to type II, Class II or MIL-PRF-62550/2 should be ordered. The latter heaters are field convertible (see 3.3.3) to the former.
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article is required (see 3.1).
e. Packaging requirements (see 5.1).

6.3 Supersession data.

Superseded Part	New Part (MIL-PRF-62550D)
M46792/2	Type I, Class I
7748716	Type I, Class II
11601809	Type I, Class II
11669489	Type II, Class II
11669489-1	Type II, Class II
11669490	Type II, Class II
11669490-1	Type II, Class II
11669496	Type I, Class II

6.4 Process average. Sampling may be initiated if the process average value for the first 100 heaters inspected is less than 1% for major defects and less than 4% for minor defects. The process average should be calculated as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of heaters inspected}} \times 100\%$$

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6.5 Subject term (key word) listing.

Air flow
Combustion vapor
Diagnostics, self
Over pressure
Smoke density
Ventilation

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

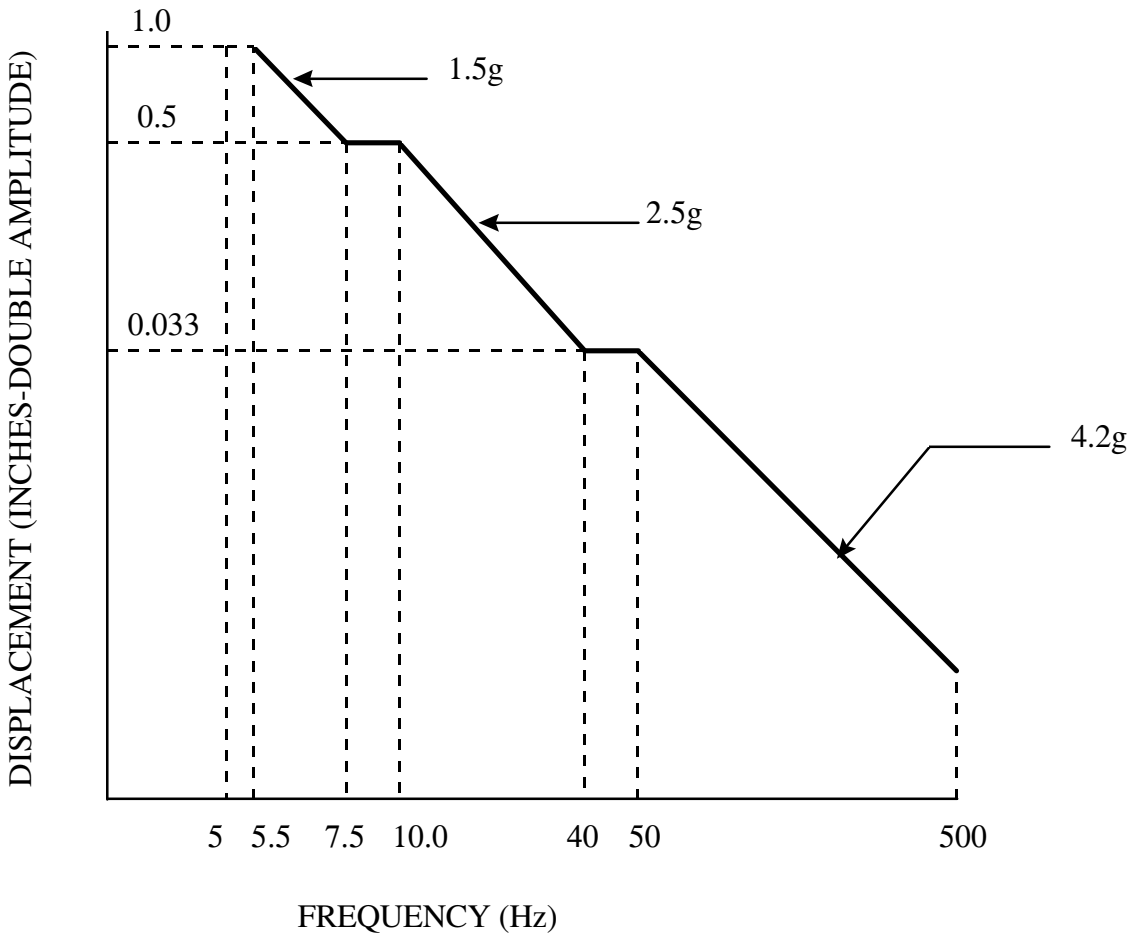


FIGURE 1. Vibration curve.

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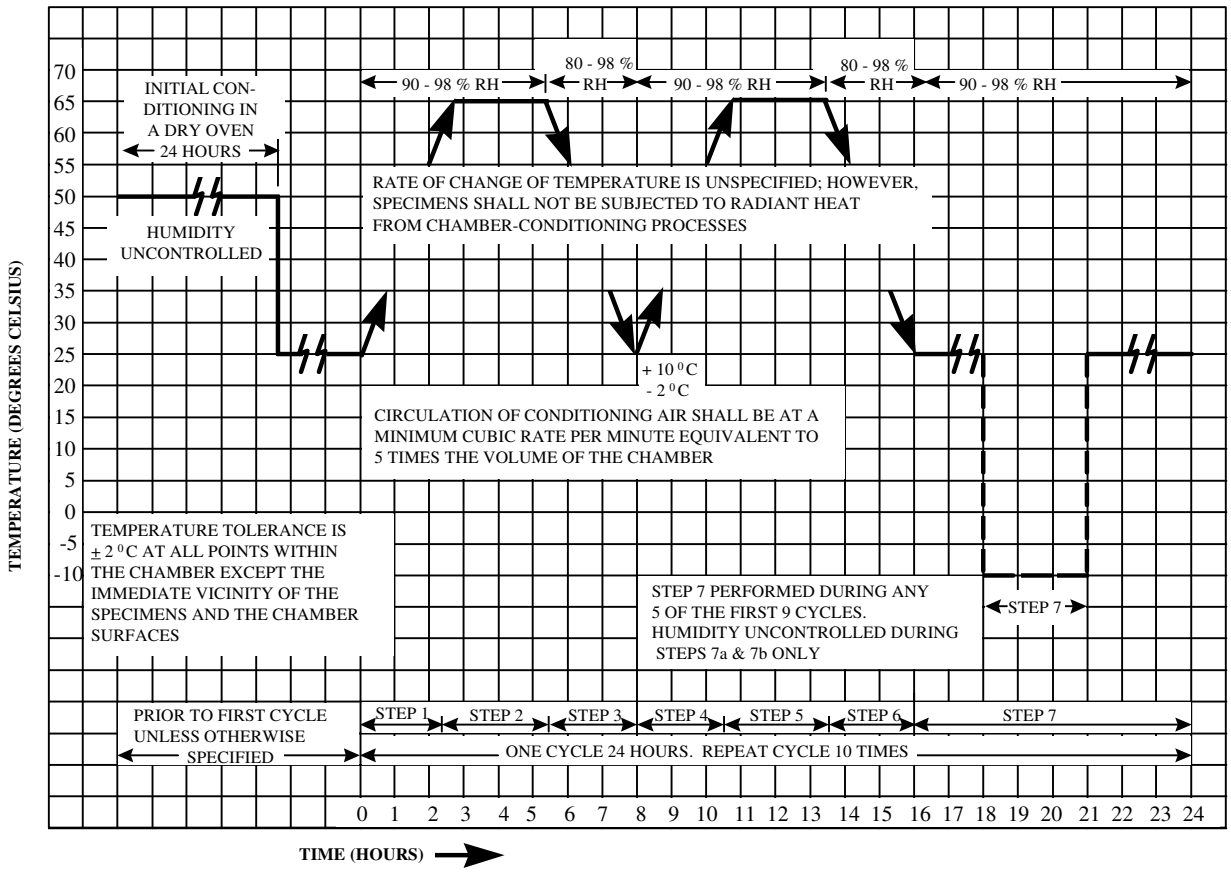


FIGURE 2. Graphical representation of moisture-resistance test.

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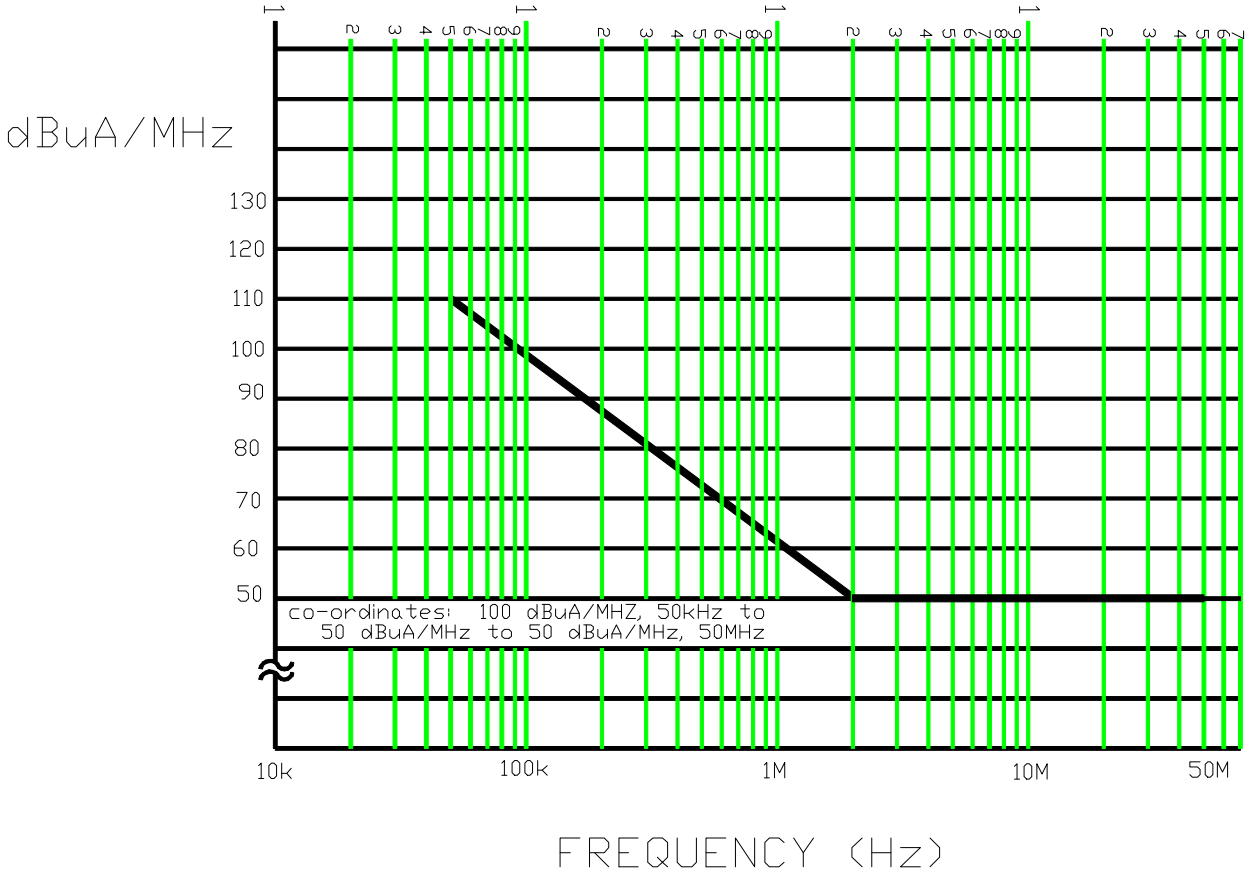


FIGURE 3a. Upper limits for conducted broadband emissions.

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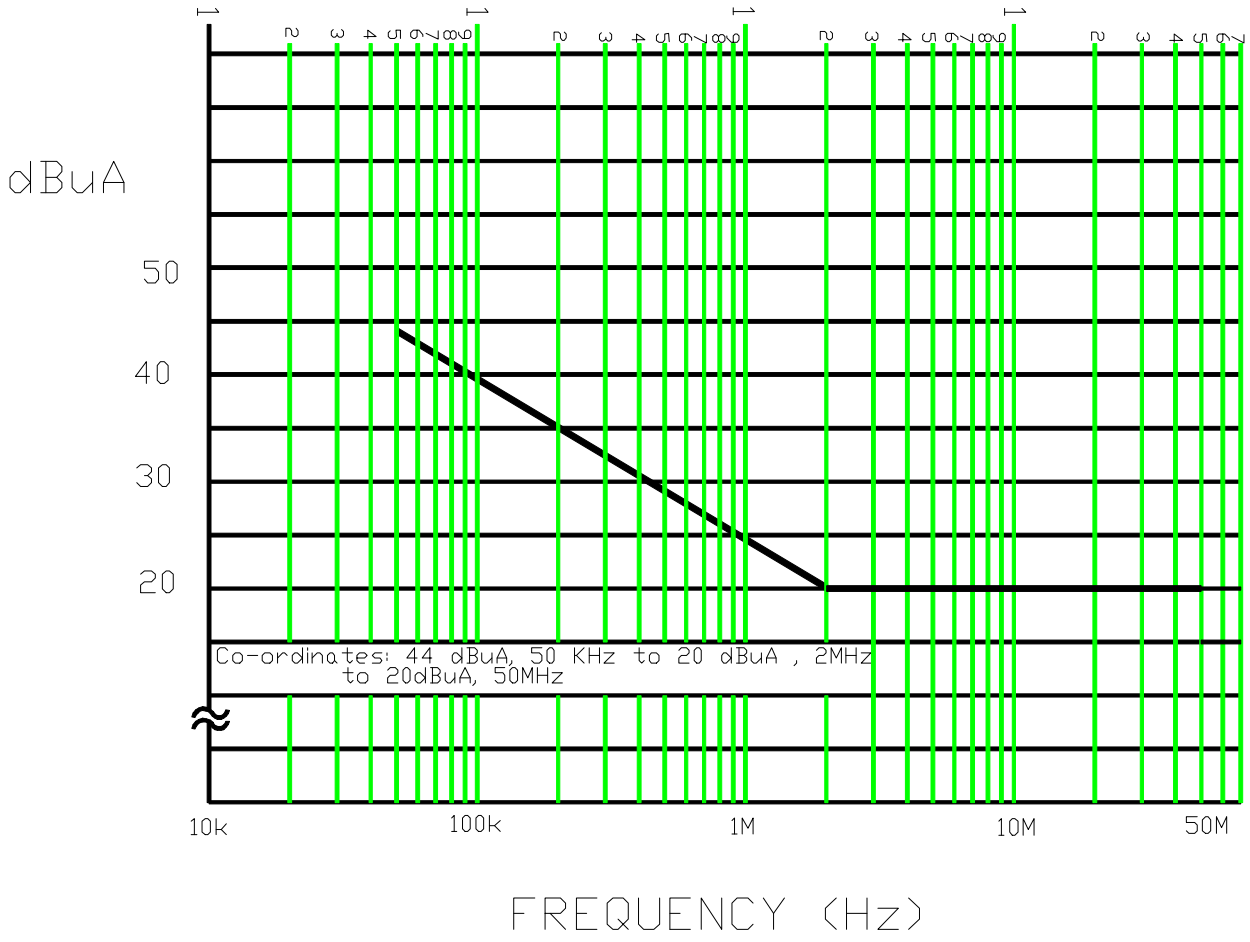


FIGURE 3b. Upper limits for conducted narrowband emissions.

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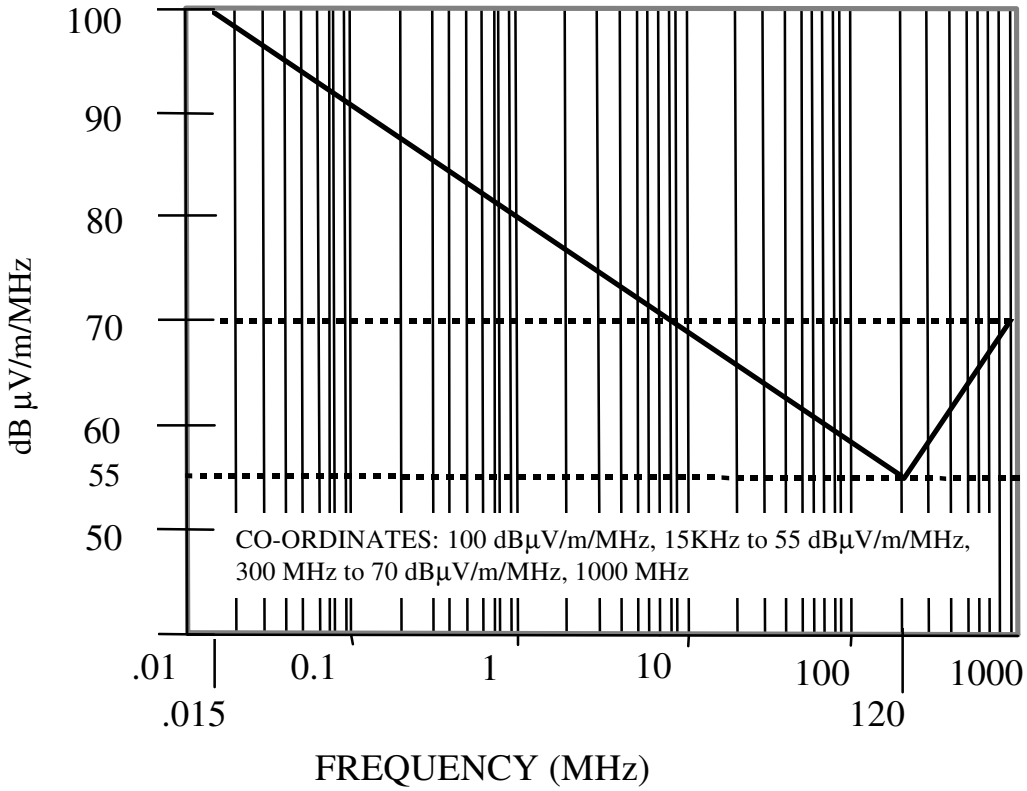


FIGURE 4. Upper limits for radiated broadband emissions.

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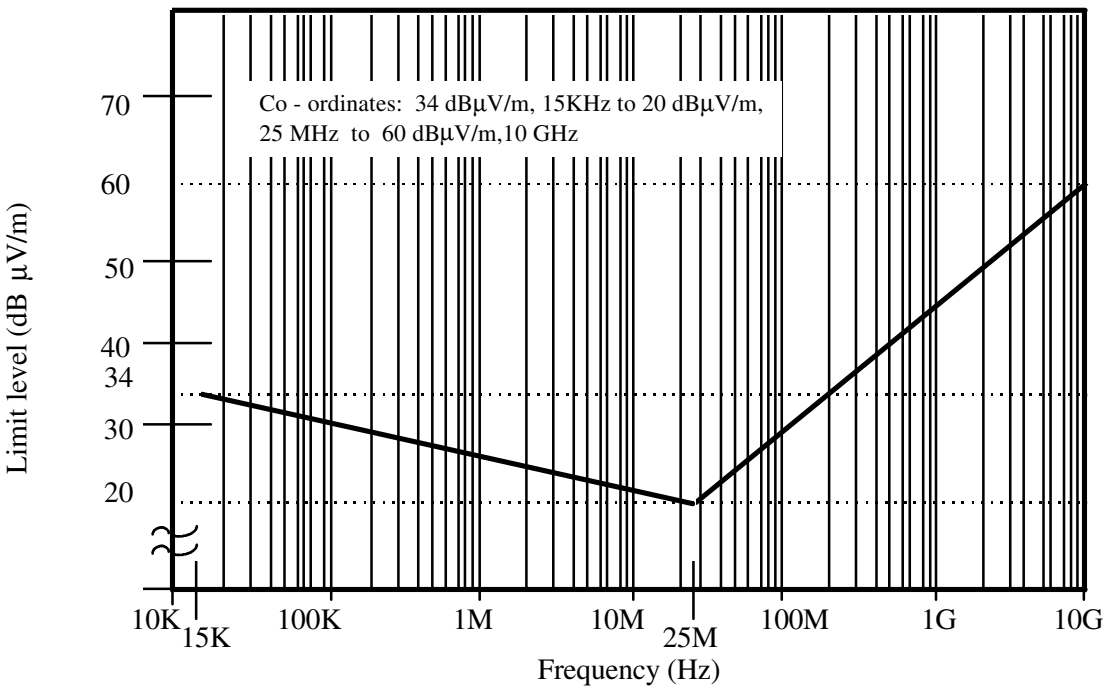


FIGURE 5. Upper limits for radiated narrowband emissions.

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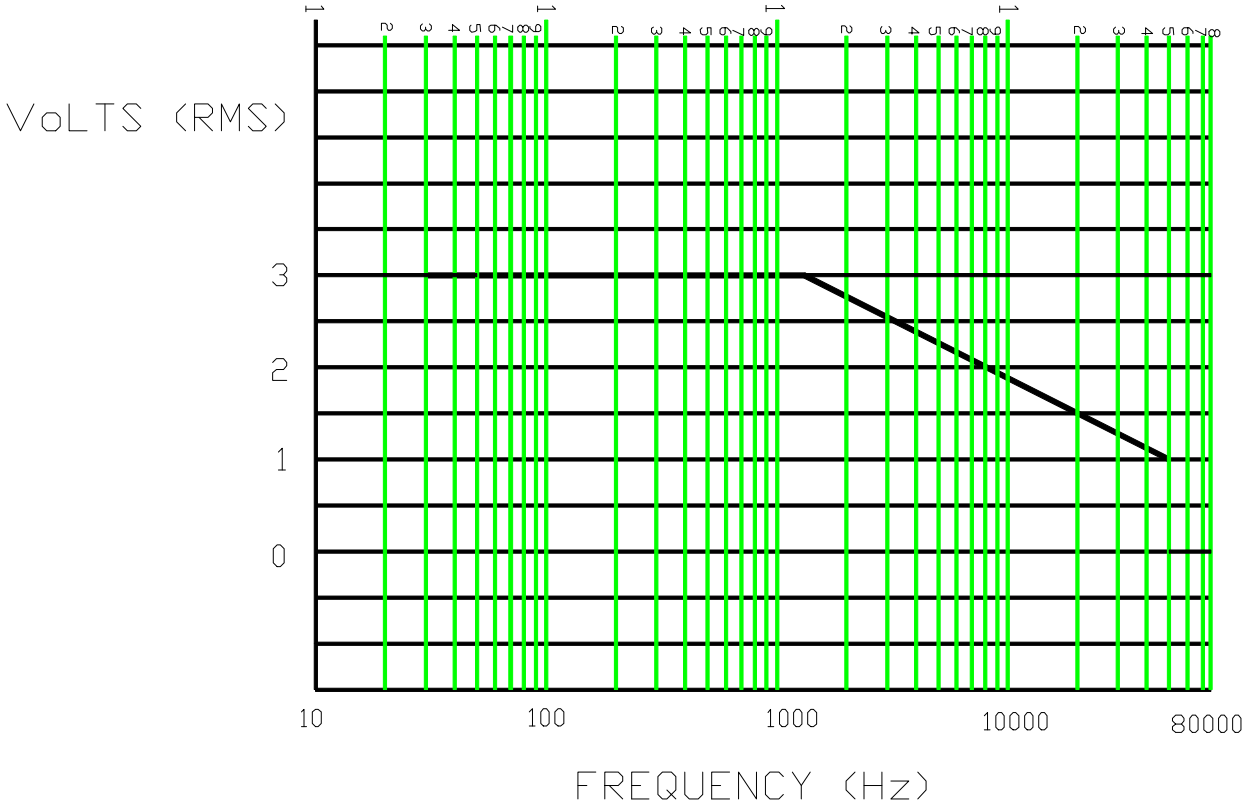


FIGURE 6. Lower limits for conducted interference signal.

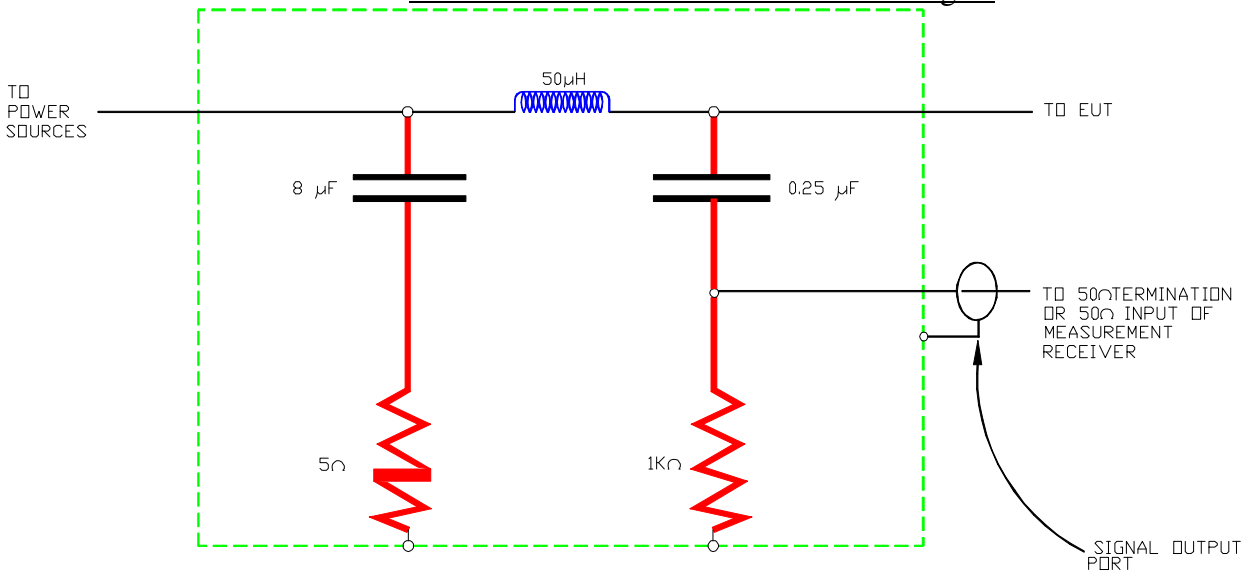


FIGURE 7. LISN schematic.

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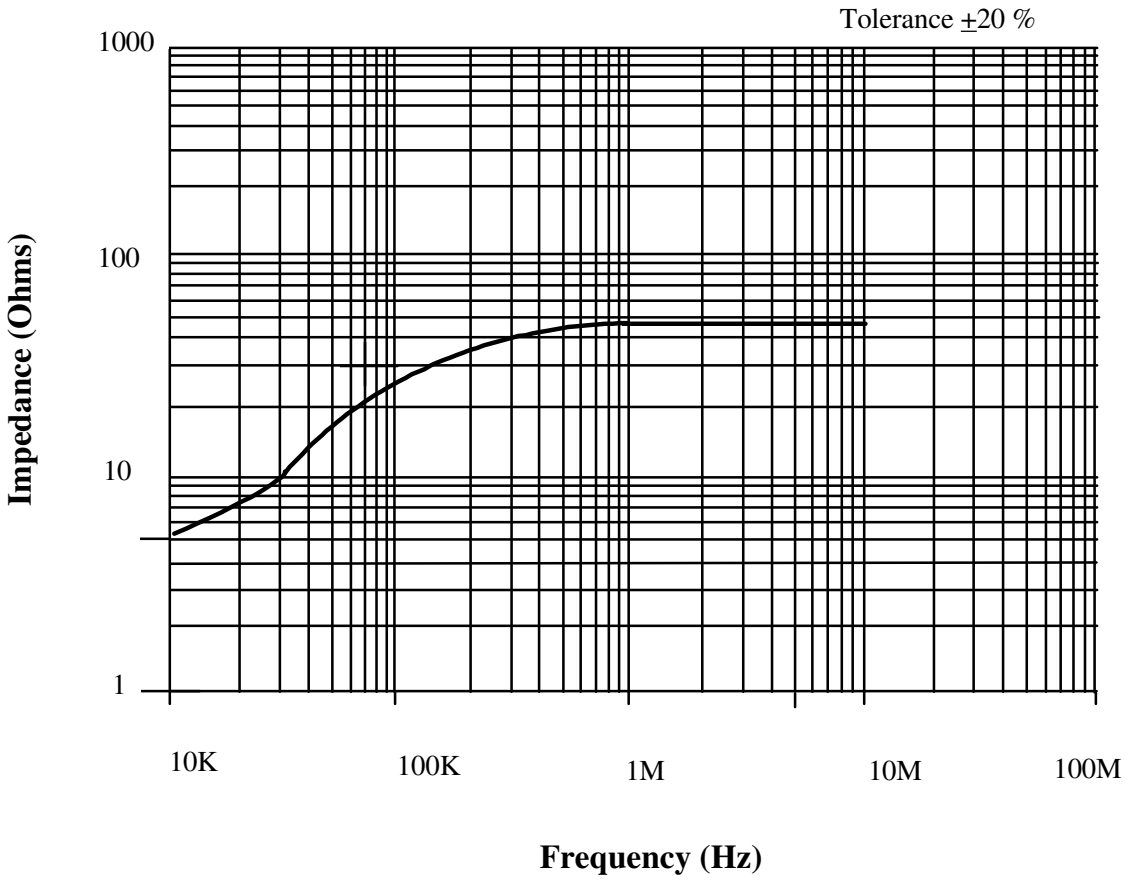


FIGURE 8. LISN impedance.

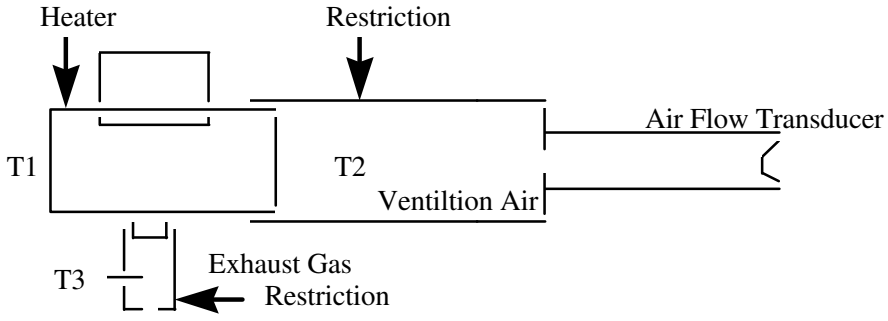


FIGURE 9. Heater set up for pressure loss testing. Seal transducer and restriction as required. Air flow measurement as specified.

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T_1 = Ambient Temperature
T_2 = Ventilation Air Temperature. 106.68 ± 2.54 cm (42 ± 1 in) from heater and within 10% of duct's center line.
T_3 = Exhaust Gas Temperature. 12.7 ± 2.54 cm (5 ± 1 in) from heater and within 10% of duct's center line.

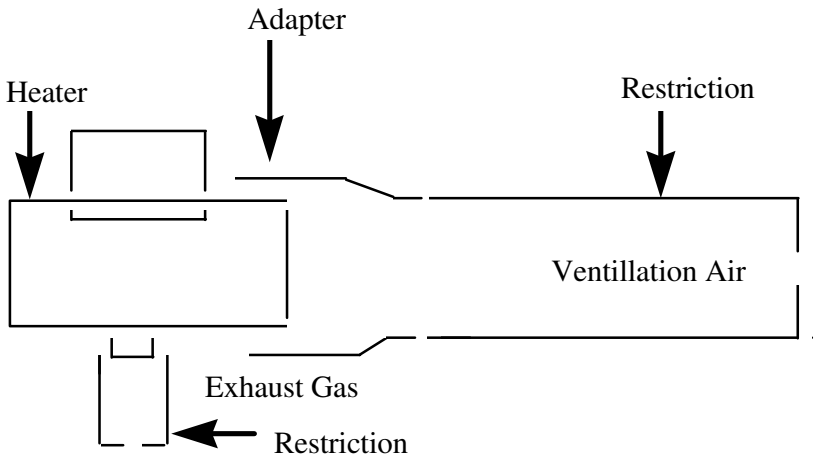


FIGURE 10. Test set up for seal leakage. Seal adapter to restriction as required. Adapter to heater dimension as specified and no sealing allowed.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2540-A042)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-62550D(AT)	**2. DOCUMENT DATE *(YYMMDD)*** **970430**

3. DOCUMENT TITLE
Heater Assembly, Combustion Vehicular Compartment, 17.6 KW (30 000/60 000 BTU/HR) General Specification for

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** (810) 574-8745 **(2) AUTOVON** 786-8745
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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